FSE10023 Linear Algebra Assignment [Semester 2 2026]

Total mark: 60

Instructions:

- Complete all 8 questions in this assignment.
- Show all solution steps clearly. Marks are awarded for both working steps and final answers.
- Submit only one assignment per group.
- Scan your completed work into a single PDF file using Adobe Scan, then upload it to Canvas.
- Ensure your submission file includes the assignment cover sheet followed by your step-by-step solutions.

Case Study: Swinburne Sarawak Campus Parking Map & Upgrades

Finding parking on campus can be tricky. Swinburne Sarawak's facilities team is planning upgrades to improve parking flow and design a new open-air parking space near campus. As a student engineering team, your task is to solve the vector math needed to map out these new parking areas.

Task 1: Basic Quantities on Campus [4 marks]

When surveying the campus parking lots, engineers measure different physical quantities.

i. Give two examples of scalar quantities related to campus parking. [2 marks]

ii. Give two examples of vector quantities related to car movement or structural forces. [2 marks]

Task 2: Finding Parking Marker Locations [14 marks]

Two survey poles, Pole A and Pole B, are placed near the multi-storey parking block. Relative to a central survey point $O(0,0)$ on the ground, their positions are $\overrightarrow{OA} = 3\hat{\imath} - 2\hat{\jmath}$ and $\overrightarrow{OB} = -2\hat{\imath} + 3\hat{\jmath}$,

i. Find the distance of Pole B from the central survey point and calculate its angle θ (where $0 \leq \theta \leq 360°$). Round θ to 1 decimal place. [6 marks]

ii. Find the direct displacement vector $\overrightarrow{OB} - \overrightarrow{OA}$ going from Pole A to Pole B. Calculate its distance and direction angle θ. [8 marks]

Task 3: Entrance Driveway Check [2 marks]

Calculate the dot product between two direction vectors $P(4,3)$ and $Q(-3,2)$ representing two intersecting driving paths near the campus entrance.

Task 4: Canopy Support Cables [3 marks]

To shade the open-air parking bays, support cables hold up a light roof structure. Two cables pull with forces $\overrightarrow{OC} = (6\hat{\imath} - 4\hat{\jmath})N$ and $\overrightarrow{OD} = (-12\hat{\imath} + 9\hat{\jmath})N$. Calculate the combined force vector using the equation $-3\overrightarrow{OC} + \frac{1}{3}\overrightarrow{OD}$. Write your final answer in unit vector form.

Task 5: Vector Cross Product Calculation [10 marks]

Given three vectors representing force directions on a support pole: $\overrightarrow{OE} = 2\hat{\imath} + 1\hat{\jmath}$, $\overrightarrow{OF} = 3\hat{\imath} - 2\hat{\jmath} + \hat{k}$ and $\overrightarrow{OG} = 3\hat{\jmath} - 4\hat{k}$. Calculate the cross product $(\overrightarrow{OE} \times \overrightarrow{OF}) \times \overrightarrow{OG}$.

Task 6: 3D Pillar Angle Calculation [14 marks]

A straight metal pillar supporting a shade structure extends from the ground point $O(0,0,0)$ to an upper joint at a point $H(4,5,7)$meters.

i. Find the direction cosines of this pillar. [8 marks]

ii. Find the angle the pillar makes with each axis. Round your answers to 1 decimal place. [6 marks]

Task 7: Pushing a Barrier Gate (Work Done) [2 marks]

A technician pushes a portable safety barrier across a straight distance path $d(6,2)$ meter by applying a force vector $F(40,30)$ Newton. Calculate the work done.

Task 8: Motorcycle Bay Surface Area [11 marks]

A small triangular plot of ground near the campus boundary is being measured for new motorcycle parking. Bounded from point $O(0,0,0)$, the two ground corners are at $P(4,1,0)$ meters and $Q(1,5,0)$ meters. Find the area of this triangular plot.